केन्द्रीय विद्यालय संगठन , कोलकाता संभाग
KENDRIYA VIDYALAYA SANGATHAN, KOLKATA REGION

प्री–बोर्ड परीक्षा/**PRE- BOARD EXAMINATION-2024-25**

कक्षा / CLASS - X अधिकतम अंक / MAX.MARKS-80

विषय / SUBECT-MATHEMATICS समय / Time : 3 Hours

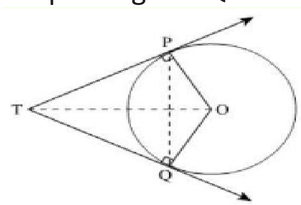
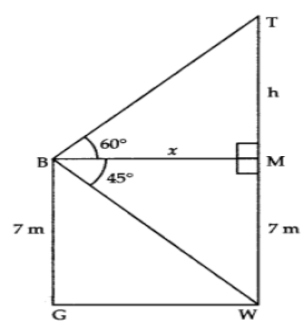
MARKING SCHEME

	Section A	
	Section A Consists of 20 questions of 1 mark each.	
1.	(b) 1/3	1
2.	(d) 1-c	1
3.	(b)15/4	1
4.	(a) a=c	1
5.	(c) 0	1
6.	(a) (2,0)	1
7.	(c)10	1
8.	(b) $a\sqrt{2}$	1
9.	(a)3/4	1
10.	(c)$2\sqrt{3}$	1
11.	(a)12/25	1
12.	(c)2	1
13.	(d)$40cm^2$	1
14.	(a) 20-25	1
15.	(d)$36cm^2$	1
16.	(b)2	1
17.	(c)1/12	1
18.	(c)8cm	1
19.	(d) Assertion (A) is false but reason (R) is true.	1
20.	(a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A)	1
	Section B	
	Section B Consists of 5 questions of 2 marks each.	
21(A).	The relationship is given by: LCM×HCF=Number 1×Number 2 64699×97=2231× Number 2 Number 2=2813	1 1
21(B).	For correct explanation $7 \times 11 \times 13 + 13$ is composite number. For correct explanation $7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$ is composite number.	1 1
22.	Let the ratio be K:1 and the point of division be (x, 0) Using Section formula between A and B with ratio K: 1 $(x,0)= (k \times 1+1 \times 4/k+1, k\times(-3)+ 1 \times 5/k+1)$	½

	On comparing $x= k+4/k+1$, $0 = -3k+5/k+1$ $-3k+5=0$ $\Rightarrow 3k=5$ $\Rightarrow k= 5/3$ Putting value of k in x, $x=\frac{\frac{5}{3}+4}{\frac{5}{3}+1} = 17/8$ Required ratio = 5: 3 and point on x-axis be (17/8,0)	½ ½ ½
23(A).	In a non-leap year, there are 52 complete weeks and 1 extra day. The extra day can be any of the seven days of the week (Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, or Saturday). The probability of having 53 Sundays in a randomly selected non-leap year is given by: P(53 Sundays)=$\frac{\text{Number of favorable outcomes}}{\text{Total outcomes}}$ $=\frac{1}{7}$	½ ½ ½ ½
23(B).	Total number of tickets is100 Total number of prize carrying tickets is 5 P(winning prize)=$\frac{\text{Number of favorable outcomes}}{\text{Total outcomes}}$ $=\frac{5}{100}$ $=\frac{1}{20}$	½ ½ ½ ½
24.	For writing distance formula. For putting correct values. For calculation and finding correct distance=$\sqrt{a^2+b^2}$	½ ½ 1
25.	For putting all correct values. For simplification and correct answer =$\frac{67}{12}$	½ 1 ½
	Section C	
	Section C Consists of 6 questions of 3 marks each.	
26.	Let the speed of the stream be x km/h. Therefore, the speed of the boat upstream = $(18 - x)$ km/h and the speed of the boat downstream = $(18 + x)$ km/h. The time taken to go upstream =$\frac{\text{distance}}{\text{speed}} = \frac{24}{18-x} h$ Similarly, the time taken to go downstream =$\frac{\text{distance}}{\text{speed}} = \frac{24}{18+x} h$ According to the question, $\frac{24}{18-x} - \frac{24}{18+x} = 1$ i.e., $24(18 + x) - 24(18 - x) = (18 - x)(18 + x)$ i.e., $x^2 + 48x - 324 = 0$ or,(x-6)(x+54)=0 or, x=6 or -54 Since x is the speed of the stream, it cannot be negative. So, we ignore the	 ½ ½ ½ 1

	root $x = -54$. Therefore, $x = 6$ Hence the speed of the stream is 6 km/h.	 ½
27(A).	(i) ΔABC ~ ΔPQR (Given) So, AB/PQ =BC/ QR= CA/ RP ……….(1) and <A =<P, <B =<Q and <C = <R ………….(2) But AB = 2 AM and PQ = 2 PN (As CM and RN are medians) So, from (1),2AM/2PN =CA/RP i.e., AM/PN=CA/RP…………….(3) Also, <MAC = <0NPR [From (2)]…………. (4) So, from (3) and (4), ΔAMC ~ ΔPNR (SAS similarity) ……………(5) (ii) From (5), CM/RN=CA/RP…………….(6) But CA/RP =AB/PQ [From (1)]……….. (7) Therefore, CM/RN =AB/PQ [From (6) and (7)] (8)	 ½ ½ ½ ½ ½ ½
27(B).	In the Δ ABC, EF ∥ BC ∴ AE/AB = AF/AC ......(1) [By BPT] In Δ ACD, FG ∥ CD ∴ AF/AC = AG/AD ....(2) [By BPT] From (1) and (2), we get AE/AB = AF/AC = AG/AD ⇒AE/AB = AG/AD ⇒$AE\times$ AD=AB×AG.	 1 1 ½ ½
28.	a) Assumes the roots of p(x) to be α and β and write the relation $\alpha = 3\beta$ The sum of the zeroes, $\alpha + \beta$ = -b/a = > $3\beta + \beta$ = -(-8)/a = > 4 β = 8/a = > β = 2/a = > $\alpha = 3\beta$ = 6/a Product of the roots, $\alpha\ \beta$ = c/a = > $6/a \times 2/a = 3/a$	 ½ ½ ½ ½

	$=> 12/a^2 = 3/a$ $=> a = 4$. (b) Since ‘a’ is positive, the graph of p(x) is open upward parabola.	 ½ ½
29.	Solution: L.H.S. = $x^2 - y^2 = (p \sec\theta + q \tan\theta)^2 - (p \tan\theta + q \sec\theta)^2$ $= p^2 \sec\theta + q^2 \tan^2\theta + 2\,pq \sec^2\theta \tan^2\theta$ $-(p^2 \tan^2\theta + q^2 \sec^2\theta + 2pq \sec\theta \tan\theta)$ $= p^2 \sec\theta + 2 \tan^2\theta + 2pq \sec\theta \tan\theta - p^2 \tan^2\theta - q^2 \sec\theta - 2pq \sec\theta \tan\theta$ $= p^2(\sec^2\theta - \tan^2\theta) - q^2(\sec^2\theta - \tan^2\theta)$ $= (p^2 - q^2)$ R.H.S. as ,$\sec^2\theta - \tan^2\theta = 1$	½ 1 ½ ½ ½
30(A).	Radius of red region=11cm. Radius of silver region include red region=11+10.5=21.5cm Area of silver region=Area of whole badge-area of red region $=\pi \times 21.5^2 - \pi \times 11^2$ $=1072.5\text{cm}^2$	½ ½ ½ ½ 1
30(B).	Central angle=60^0 To find the area of one equilateral triangle=333.2cm^2 To find the area of Area of one sector= $\frac{1232}{3}\text{cm}^2$ To find the area of Area of 6 design=$464.8\ \text{cm}^2$ Cost of making $464.76\ \text{cm}^2$ designs = 464.8×0.35 = Rs 162.68	 1 1 ½ ½
31.	For correct proof.	3
	Section D	
	Section D Consists of 4 questions of 5 marks each.	
32(A).	Let unit’s place digit be x and ten’s place digit be y. ∴ Original number = x + 10y Reversed number = 10x + y According to the Question, 10x + y = 3(x + 10y) – 9 ⇒ 10x + y = 3x + 30y – 9 ⇒ 10x + y – 3x – 30y = -9 ⇒ 7x – 29y = -9 …………….(i) Again, 10x + y – (x + 10y) = 45 ⇒ 9x – 9y = 45 ⇒ x – y = 5 …[Dividing both sides by 9]…………...(ii) Now, multiplying equation (ii) by 7, we get 7x – 7y = 35…………….(iii) Subtracting Equation (i) from equation (iii), –7y + 29y = 35 + 9 ⇒ 22y = 44 ⇒ y = 2 Putting the value of y in (ii), x = 5 + 2 = 7	 ½ ½ 1 ½ 1 ½ ½

	∴ Original number = x + 10y = 7 + 10(2) = 27	½
32(B).	For drawing correct graph For finding correct solution x=2 and y=3 graphically.	3 2
33.	To prove the lengths of tangents drawn from an external point to a circle are equal. For proving<PTQ = 2<OPQ.  We have TP=TQ In ΔTPQ,TP = TQ ⇒∠TQP=∠TPQ……(1) (In a triangle, equal sides have equal angles opposite to them) ∠TQP+∠TPQ+∠PTQ=180∘ (Angle sum property) ∴2∠TPQ+∠PTQ=180∘ (Using (1)) ⇒∠PTQ=180∘−2∠TPQ……(1) OP⊥PT ∴∠OPT=90∘ ⇒∠OPQ+∠TPQ=90∘ ⇒∠OPQ=90∘−∠TPQ ⇒2∠OPQ=2(90∘−∠TPQ) =180∘−2∠TPQ……(2) From (1) and (2), we get ∠PTQ=2∠OPQ	3 ½ ½ ½ ½
34.	Let BG be building TW be Tower, then: BM = x, ∠MBT = 60° ∠MBW = 45° Draw BM ± TW In rt. ∠d ΔBMW tan 45° = WM/BM ⇒ 1 = 7x ⇒ x = 7 m  In rt. ∠d ΔTMB tan 60° = TM/BM ⇒ √3 = h/x ⇒ h = √3x = 7√3 Height of Tower = TW = TM + MW = (7√3 + 7)m = 7(√3 + 1)m	Diagram (1) 1 1 1 1

35(A).

Class interval:	Frequency: (f_i)	Cumulative frequency ($c.f.$)
0-10	f_1	f_1
10-20	5	$5+f_1$
20-30	9	$14+f_1$
30-40	12	$26+f_1$
40-50	f_2	$26+f_1+f_2$
50-60	3	$29+f_1+f_2$
60-70	2	$31+f_1+f_2$
	$N=40=31+f_1+f_2$	

1

Here $31+f_1+f_2=40$

i.e. $f_1+f_2=9$………………(1) ½

Since median = 32.5 so the median class is . $30-40$ ½

Here, $l=30, f=12, F=14+f_1$ and $h=10$

$$\text{Median}=l+\left\{\frac{\frac{N}{2}-F}{f}\right\}\times h$$ ½

$$32.5=30+\left\{\frac{20-(14+f_1)}{12}\right\}\times 10$$ ½

$$2.5=\frac{(6-f_1)\times 10}{12}$$

$$2.5\times 12=60-10f_1$$

$$10f_1=60+30$$

$$f_1=\frac{30}{10}$$

$$=3$$ 1½

Putting the value of f_1 in (1), we get

$$f_2=9-3$$ ½

$$=6$$

Hence, the missing frequencies are 3 and 6.

35(B).

To find mean = 69.6(by any suitable method). 2 ½

To find mode = 76.67 2 ½

Section E

36.

(i) A.P .is 5,8,11,14,… 1

Common difference=3

(ii) a $_7$=a+6d 1

	$=5+6*3$ $=23$	
(iii)	(A) Total distance covered by the competitor = 10 m + 16 m + 22 m + ... up to 10 terms $=10/2[2\times10+(10-1)\times6]$ $=5\times(20+54)$ $=5\times74$ $=370$m (B) Total distance covered by the competitor = 10 m + 16 m + 22 m + ... up to 12 terms $=12/2[2\times10+(12-1)\times6]$ $=6\times(20+66)$ $=6\times86$ $=516$m	2 2
37.		
(i)	In ΔCAB and ΔCLO, we have <CAB = <CLO 90° <C = <C (common) . By AA similarity criterion, ΔCAB ~ ΔCLO	1
(ii)	In ΔDCA and ΔBAC, DC = BA [x = y (Given)] <DCA = <BAC CA = AC [Each 90°) [Common] By SAS similarity criterian, ΔDCA - ΔBAC So BC/DA=DC/BA BC: DA = 1 : 1	1
(iii)(A)	if CL = a $\Rightarrow a/z = d/x$ $a = zd/x$	2
(iii)(B)	if AL = b $\Rightarrow b/z = d/y$ $b = zd/y$	2
38.		
(i)	Given, edge of each cubical box (a) = 20 cm Area to be painted = Area of 14 square faces $= 14\ a^2$ $= 14\ (20)^2$	1

	$= 5600\ cm^2$	
(ii)	From figure, Length of the pot (l) = 20 cm Breadth of the pot (b) = 20 x 4 = 80 cm and height of the pot (h) = 20 cm Volume of pot = lbh = 20 x 80 x 20 $= 32000\ cm^3$	1
(iii)	(A) Height of sapling=11cm Diameter=14.2cm Radius=7.1cm Volume of 1 sapling=$1742.75 cm^3$ OR (B) Total volume of pot = $32000\ cm^3$ Volume of 4 saplings = 1742.75 x 4 = $6971\ cm^3$ Volume of compost and soil =32000 - 6971 =$25029\ cm^3$	2 2